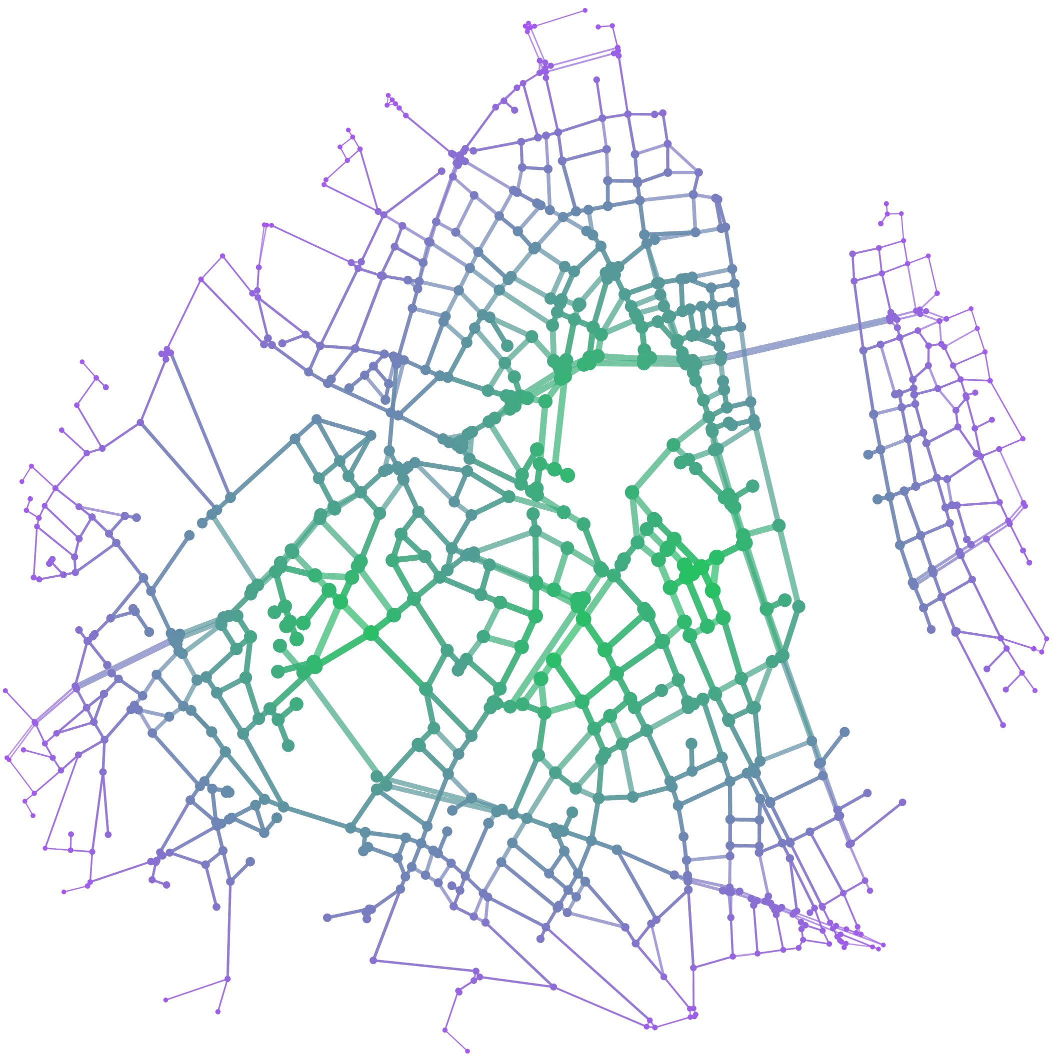
MEB

Mathematical Essays from Bonn



Bonn, 2026.1

Essays

a very long and nice title

Mathematical Essays Bonn

Hallo ich werde ein sample essay. Ich zitiere [1]. Darin steht nichts, denn es gibt dieses Buch gar nicht.
Hello, I am going to be a sample essay! Ich zitiere [1]. Nothing is written in there, because the book doesn't exist.

> **Bemerkung 1.** Diese Bemerkung begrüßt dich auf Deutsch.

> **Remark 2.** Diese Bemerkung begrüßt dich auf Englisch.

In der Bemerkung 2 wird man auf Englisch begrüßt. Unverständlich!
You can even link to other webpages! `www.google.com`

> **Theorem 3** (with a big name)
>
> hello

> **Definition 4**
>
> hello

> **Beispiel 5.** guten Tag! Ich bin ein Beispiel

> **Example 6.** hello

Figure 1 is great!

Beweis. Das ist ein Beweis auf deutsch! □

Literatur

[1] MEB. *MEBs first book.* MEB, 2026.

MEB ist eine wunderbare Essaykollektive, um deinen mathematischen Gedanken freien Lauf zu lassen. MEB gibt es seit dem Wintersemester 2025/2026.

Proof. This is an english proof. The claim is correct! □

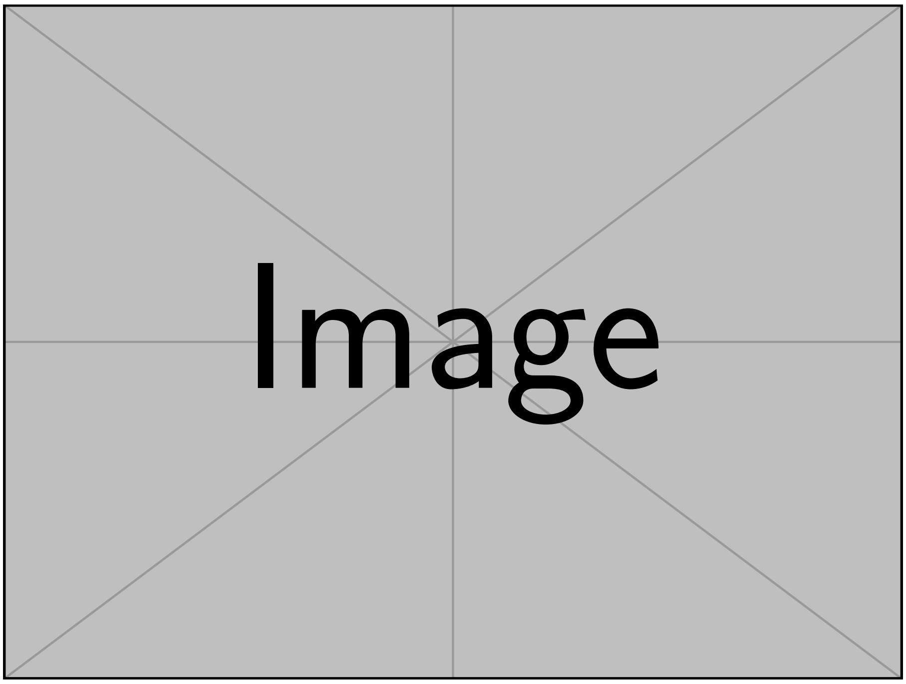


Abbildung 1: This is a figure.

Strengths and Limits of Formalisation: CW complexes as an example

Hannah Scholz

Mathematical formalisation is an area of mathematics that has seen exponential growth in the last few years. Simplified, formalisation is the practice of writing a proof in a programming language and then using the computer to verify the validity of the argument.
Essentially, this works because a proof that A implies B can be seen as a function that sends a proof of A to a proof of B. So when you write a proof of "A implies B" in a programming language, you are specifying such a function. Checking the validity of the proof then corresponds to checking that this function indeed sends proofs of A to proofs of B. This correspondence between proofs and programs is called the Curry-Howard isomorphism.
With the advent of powerful AI tools, formalisation has come to be seen as a magic tool that can lend absolute credibility to anyone using it. Every day we see new claims of solutions to previously unsolved problems supposedly backed by formalisation. Therefore, it is critical that we understand what formalisation can actually be helpful with and where it's limits lie.
This is the focus of the following essay. I will explore this topic with my formalisation of CW complexes in the language Lean as an example. This text assumes no prior knowledge of formalisation or CW complexes. However, I will not explain basic terms from topology but the text should still be understandable without knowing them. This essay is heavily inspired by the essay "Why formalize mathematics?" written by Patrick Massot [1]. So if this topic interests you, please check out that essay as well!

What is a CW complexes?

CW complexes are an important classes of spaces from the area of topology. Their importance is owed to the fact that they are very general spaces that at the same time allow for very powerful results. Intuitively, a CW complex is the result of glueing n-cells together along their edges where an n-cell is a continuous image of an n-disc. You can see examples of n-cells of different dimensions in Figure 1.
By gluing these cells together you can build a lot of the spaces that you already know. In Figure 2, you can see that intervals, the real line and spheres are all CW complexes.
Now we should move from this intuition to the concrete definition. I know, it looks very long and scary but don't worry: I will offer an intuitive explanation below and you don't need to understand all of the details.

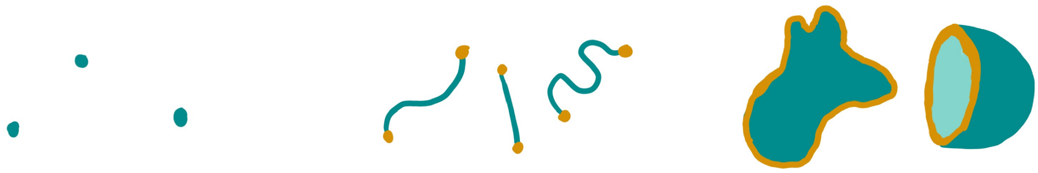


Figure 1: 0-, 1- and 2-cells.

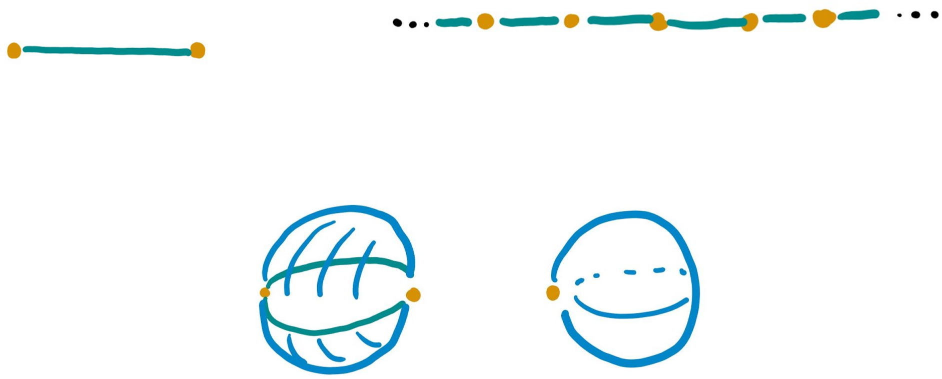


Figure 2: CW structures on the interval, real line and two structures on the 2-sphere.

Definition 1

Let X be a Hausdorff space. An *(absolute) CW complex* on X consists of a family of indexing sets $(I_n)_{n\in\mathbb{N}}$ and a family of continuous maps $(Q_i^n\colon D^n \to X)_{n\in\mathbb{N}, i\in I_n}$ called *characteristic maps* with the following properties:

(i) $Q_i^n|_{\mathrm{int}(D^n)} : \mathrm{int}(D^n) \to Q_i^n(\mathrm{int}(D^n))$ is a homeomorphism for every $n \in \mathbb{N}$ and $i \in I_n$. We call $e_i^n := Q_i^n(\mathrm{int}(D^n))$ an *(open) n-cell* and $\overline{e}_i^n := Q_i^n(D^n)$ a *closed n-cell.*

(ii) Two different open cells are disjoint.

(iii) For each $n \in \mathbb{N}$ and $i \in I_n$ the *cell frontier* $\partial e_i^n := Q_i^n(\partial D^n)$ is contained in the union of a finite number of closed cells of a lower dimension.

(iv) A set $A \subseteq X$ is closed if the intersections $A \cap \overline{e}_i^n$ are closed for all $n \in \mathbb{N}$ and $i \in I_n$.

(v) The union of all closed cells is X.

Here D^n refers to the closed unit n-disc. The convention for what the symbol D means is different in topology to other fields of mathematics.

Now for a short explanation: property (i) tells you that the open disk needs to be deformed in a reasonable way (e.g. no ripping or gluing). Note that the notation for the closed n-cell $\overline{e}_i^n$ is very confusing: it looks like the closed cell is the closure of the open cell but this in not the case in general. Property (iii) is called "closure finiteness" (which is the "C" in "CW complex"). It tells you that the cell frontier needs to be glued to cells of lower dimension and the new cell cannot stretch over infinitely many lower cells. Again, the notation for the cell frontier is confusing: it is not generally the frontier of the open cell. The "W" in "CW complex" stands for "weak topology". This is Property (iv). It tells you that the topology on X is determined by the cells. Lastly, Property (v) tells you that the whole space is made up of the cells.

Strength 1

Definitions and notation are unambiguous.

References

[1] Patrick Massot. *Why formalize mathematics?* 2021.

Here, you can write a few(!) third-person sentences about yourself, your interests and if you want to, a link to your website or similar. If you don't want a blurb to appear, put a % before the command (without the backslash).

Contributors

Authors

Axioml, Testautor Bert, Testautor Ernie

Editors

LaTeXand Design

You?

If you are a math person from Bonn and would like to help with the next edition of MEP, you can write us an e-mail at gpohlenz@uni-bonn.de. If you want to be added to our WhatsApp community, include your phone number. To get started with writing your essay, clone our repository https://github.com/MEB-Collective/MEB.